

ORIGINAL RESEARCH

Nonfatal Adverse Events and Risk for Subsequent Mortality in Patients Undergoing Percutaneous Coronary Intervention

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ABSTRACT

BACKGROUND Current guidelines and quality metrics after percutaneous coronary intervention (PCI) often assign equal weight to nonfatal adverse events, including heart failure hospitalization (HFH), acute coronary syndrome (ACS), and major bleeding, within composite endpoints.

OBJECTIVES The aim of this study was to quantify and compare the associations of HFH, ACS, and major bleeding with subsequent mortality following PCI.

METHODS A Japanese multicenter prospective PCI registry (2008–2021) with 2-year postprocedural outcomes was analyzed. The primary outcome was all-cause mortality, with HFH, ACS, and major bleeding as time-varying exposures. Cumulative incidence was estimated with death as a competing risk. Cox proportional hazards models and population attributable fractions were used to assess associations between time-varying events and mortality, adjusting for conventional risk factors.

RESULTS Among 10,482 patients (mean age 69 years, 77.5% men) followed for a median of 730 days (Q1–Q3: 730–730 days), 1,021 (9.7%; 95% CI: 9.2%–10.3%) experienced adverse events. Two-year cumulative incidence rates were 4.7% (95% CI: 4.2%–5.1%) for HFH, 3.4% (95% CI: 3.0%–3.7%) for ACS, and 2.5% (95% CI: 2.2%–2.8%) for major bleeding. Cox analysis showed that HFH was associated with mortality (adjusted HR [aHR]: 6.11; 95% CI: 4.76–7.85), followed by ACS (aHR: 3.22; 95% CI: 2.14–4.84) and major bleeding (aHR: 2.62; 95% CI: 1.71–4.02). Population attributable fraction analysis demonstrated that HFH accounted for 20.1% (95% CI: 19.0%–21.0%) of mortality burden, higher than ACS (4.3%; 95% CI: 3.3%–4.9%) and major bleeding (2.9%; 95% CI: 1.9%–3.5%).

CONCLUSIONS HFH shows stronger associations with mortality than ACS or major bleeding following PCI, suggesting that adverse events should not be weighted equally and underscoring the need to reconsider prioritization in clinical decision-making and endpoint definitions. (JACC Cardiovasc Interv. 2026;■:■–■) © 2026 by the American College of Cardiology Foundation.

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**ABBREVIATIONS
AND ACRONYMS****ACC** = American College of
Cardiology**ACS** = acute coronary
syndrome(s)**aHR** = adjusted HR**CAD** = coronary artery disease**CCS** = chronic coronary
syndrome**HFH** = heart failure
hospitalization**NCDR** = National
Cardiovascular Data Registry**PAF** = population attributable
fraction**PCI** = percutaneous coronary
intervention

Contemporary clinical guidelines and quality metrics after percutaneous coronary intervention (PCI) commonly incorporate nonfatal adverse events (eg, heart failure hospitalization [HFH], acute coronary syndrome [ACS], and major bleeding) into composite endpoints while implicitly conferring equivalent clinical significance to each component.^{1,2} Although this simplification facilitates trial design and benchmarking, it may obscure important heterogeneity in their associations with subsequent all-cause mortality.

Previous studies, including landmark randomized controlled trials, have generally assessed these complications in isolation. Such compartmentalization precludes an integrated appraisal of the cumulative burden

and potential interactions among multiple adverse events within a single cohort.^{3,4} Furthermore, recent trials such as ISCHEMIA (International Study of Comparative Health Effectiveness With Medical and Invasive Approaches) and PARADISE-MI (Prospective ARNI vs ACE Inhibitor Trial to Determine Superiority in Reducing Heart Failure Events After MI) have shown that residual vulnerability in PCI-treated patients extends beyond recurrent ACS and bleeding and increasingly implicates HFH as a determinant of long-term prognosis.⁵⁻⁷

Despite these observations, the comparative impact of HFH, ACS, and major bleeding on mortality has not been quantified within a unified analytical framework. Consequently, empirical guidance is lacking on how these events should be weighted in composite endpoints and win-ratio analyses, and clinicians are unable to determine which complication should receive the highest priority in surveillance and secondary prevention. A clearer delineation of their differential contributions could refine risk stratification and optimize management after PCI.

Therefore, we analyzed data from a prospective registry encompassing multiple centers in Japan with 2-year follow-up. Using time-dependent Cox models and population attributable fraction (PAF) estimation, we quantified and contrasted the impact of HFH, ACS, and major bleeding on subsequent mortality. By quantifying the mortality burden attributable to HFH, ACS, and major bleeding within a single prospective cohort, we aimed to refine post-PCI prognostic models, direct surveillance, and preventive interventions toward the complications with the greatest lethality and to supply empirically derived

weights for constructing composite and win-ratio endpoints in future cardiovascular trials.

METHODS

STUDY SETTING AND POPULATION. We obtained data from the JCD-KiCS (Japan Cardiovascular Database-Keio Inter-Hospital Cardiovascular Studies) registry. JCD-KiCS is an ongoing, prospective, multicenter registry including 15 tertiary and secondary hospitals in the Tokyo metropolitan area and surrounding prefectures that was established in alignment with the American College of Cardiology's (ACC) National Cardiovascular Data Registry (NCDR), using universal definitions for both variables and outcomes.^{8,9} The data collection framework was structured following the ACC NCDR CathPCI Registry protocol (version 4.1.2), ensuring standardized variable definitions and reporting methods.^{10,11}

A unique strength of JCD-KiCS, in contrast to the ACC NCDR framework, is that this registry independently captures 2-year clinical outcomes of its initial registrants, thereby providing valuable long-term follow-up data. To achieve comprehensive follow-up of clinical outcomes, we implemented a 3-pronged approach to data collection at 1 and 2 years after index PCI: 1) reviewing PCI hospital medical records; 2) conducting telephone interviews with patients; and 3) sending letters to patients or their primary care physicians. The collected data encompassed all-cause mortality and nonfatal cardiovascular events.

The study was approved by the Institutional Review Board committee of Keio University (20080073) and the respective Institutional Review Boards of the participating hospitals. The study population consisted of consecutive patients who underwent PCI between September 2008 and October 2021 and survived their index hospitalizations for PCI. Patients were excluded if data regarding their primary indications for PCI (ACS or chronic coronary syndrome [CCS]) was unavailable. All remaining patients were incorporated into our analytical cohort.

OUTCOME MEASURES AND EXPOSURE DEFINITIONS.

The primary outcome was all-cause mortality. The exposures of interest were post-PCI nonfatal adverse events, including HFH, major bleeding, and ACS. The diagnosis of HFH required the presence of at least 1 characteristic symptom (such as dyspnea or edema) along with a minimum of 2 confirmatory findings, including physical examination (bilateral rales, elevated venous pressure), diagnostic imaging

(interstitial or alveolar edema on chest radiography), or the administration of heart failure-specific interventions (intravenous diuretic agents or inotropic medications). Patients who developed heart failure symptoms during hospitalization for myocardial infarction were not classified as HFH, as myocardial infarction was considered the primary diagnosis in these cases. On the basis of findings that Bleeding Academic Research Consortium type ≥ 2 bleeding is associated with a significant increase in 1-year mortality, major bleeding in this study was defined as bleeding equivalent to Bleeding Academic Research Consortium type 2 or higher, requiring hospitalization or more intensive treatment.^{12,13} Patients with ACS were those with PCI indications of ST-segment elevation myocardial infarction, non-ST-segment elevation myocardial infarction, or unstable angina. The diagnosis of myocardial infarction was established on the basis of criteria specified in the fourth universal definition of myocardial infarction.¹⁴

STATISTICAL ANALYSIS. Continuous variables are expressed as mean $\pm$ SD or median (Q1-Q3) and categorical variables as counts and percentages. Baseline covariate distributions are presented separately for patients with ACS and CCS. Additionally, patient characteristics were summarized for each post-PCI event (HFH, major bleeding, and ACS).

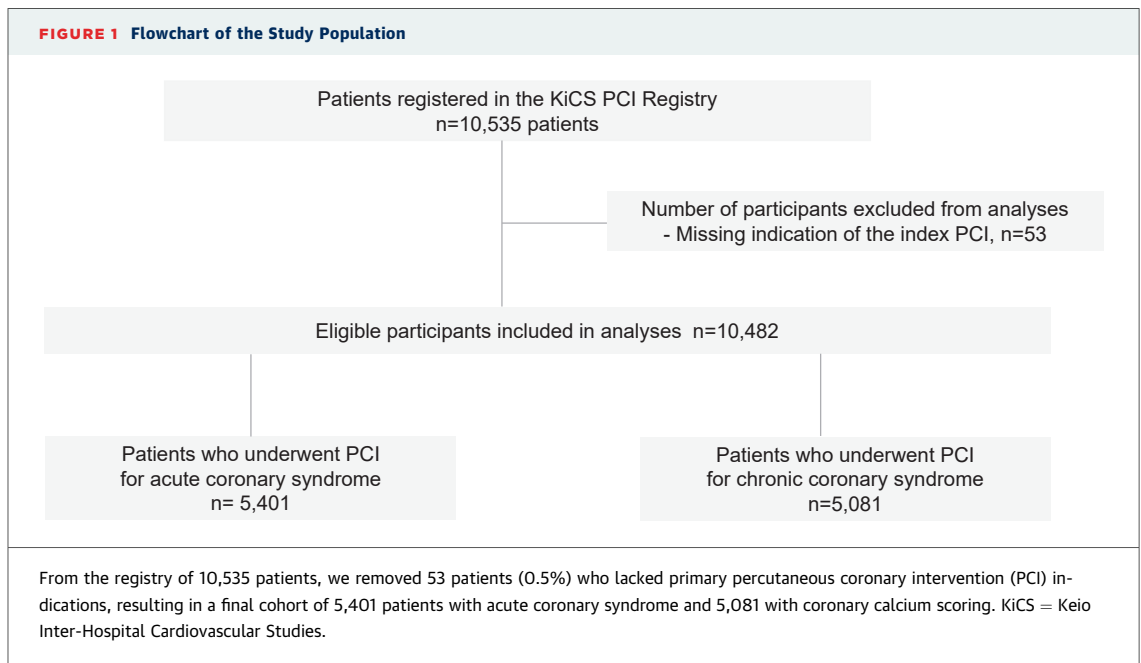
First, the frequency of each event was visualized using a Venn diagram. To estimate the cumulative incidence of each nonfatal event (HFH, major bleeding, and ACS) within 2 years after PCI, we used the cumulative incidence function based on the Aalen-Johansen estimator, treating death as a competing risk and censoring patients at 2 years of follow-up.

Then, we used frailty Cox proportional hazards models with time-varying covariates, accounting for clustering within hospitals, to examine the association between each clinical post-PCI event and subsequent mortality (in separate analyses). Specifically, post-PCI events were modeled as time-varying exposures: patients were considered unexposed until the date of the incident adverse event, after which their status was updated to exposed for the remaining follow-up period. The associations between post-PCI events and mortality were assessed using Cox proportional hazards regression to estimate unadjusted (model 0) and adjusted HRs (aHRs). Model 1 adjusted for age and sex. Model 2 further adjusted for body mass index, estimated glomerular filtration rate, smoking status, hypertension, diabetes mellitus, dyslipidemia, previous myocardial infarction, previous heart failure, heart failure symptoms at index admission, peripheral vascular disease,

cerebrovascular disease, previous PCI, and the indication for the index PCI. The fully adjusted model (model 3) further post-PCI medications. Covariates were selected a priori on the basis of clinical plausibility and established prognostic factors identified in previous studies.^{9,15} To evaluate whether the combined impact of multiple adverse events on mortality was synergistic rather than additive, we tested for 2-way interaction terms among the time-varying covariates (HFH, ACS, and major bleeding) in model 3.

Stratified analyses were performed by 1) initial presentation (ACS vs CCS); and 2) history of heart failure (previous heart failure or heart failure symptoms at index admission), including interaction tests between adverse events and these stratifying variables. To address immortal time bias, we conducted a sensitivity analysis examining the association between events occurring within the first 180 days post-PCI and subsequent 1-year mortality. Furthermore, to assess the independent prognostic impact of each event within a unified framework, we performed an additional sensitivity analysis using a multivariable Cox proportional hazards model that included HFH, ACS, and major bleeding simultaneously as time-varying covariates. We examined the proportionality assumption for baseline covariates using Schoenfeld residuals and found no instances in which the assumption was violated. Primary analyses were performed as complete-case analyses; missingness patterns were evaluated ([Supplemental Table 1](#)), and because Little's test suggested that data were not missing completely at random ($P < 0.05$), we conducted a sensitivity analysis using multiple imputation by chained equations (20 imputations) for model 3 to assess robustness.

We also calculated PAFs using model 3 to quantify the proportion of subsequent mortality attributable to the 3 adverse events (HFH, ACS, and major bleeding) analyzed in our study. This epidemiologic approach enabled us to estimate the relative contribution of each event type to the overall mortality burden among post-PCI patients.¹⁶ Additionally, to evaluate the specific associations with cardiac-related outcomes, we performed a supplemental PAF analysis with cardiovascular death as the outcome. Cardiovascular death was defined as death due to myocardial infarction, heart failure, sudden cardiac death, stroke, or other cardiovascular causes, as adjudicated by the event committee. Analyses were performed using R version 4.2.3 (R Foundation for Statistical Computing), and 2-sided P values <0.05 were considered to indicate statistical significance.



RESULTS

PATIENT POPULATION. Figure 1 shows the study inclusion and exclusion flowchart. Among 10,535 patients registered with 2-year outcomes, we excluded 53 patients (0.5%) missing information on their primary indications for PCI (ACS or CCS). This resulted in a final cohort of 10,482 patients (99.5%) (mean age 69 years, 77.5% men). The median follow-up duration was 730 days (Q1-Q3: 730-730 days), with a total of 7,285,682 person-days of follow-up. The baseline characteristics and procedural data, stratified by the primary indication for PCI, are summarized in Table 1 and Supplemental Tables 1 and 2.

INCIDENCE OF NONFATAL ADVERSE EVENTS. During a median follow-up period of 2 years, 481 of 10,482 patients (4.6%) developed HFH, followed by 353 of 10,482 patients (3.4%) with ACS and 259 of 10,482 patients (2.5%) with major bleeding (Supplemental Figure 1). The median time to event was 183 days (Q1-Q3: 57-409 days) for HFH, 237 days (Q1-Q3: 95-460 days) for ACS, and 249 days (Q1-Q3: 105.5-464.5 days) for major bleeding. Regarding the primary outcome, the median time to death among nonsurvivors was 306 days (Q1-Q3: 147-503 days). When accounting for death as a competing risk, in the overall cohort, the 2-year cumulative incidence of post-PCI clinical events was 4.7% (95% CI: 4.2%-5.1%) for HFH, 3.4% (95% CI: 3.0%-3.7%) for ACS, and 2.5% (95% CI: 2.2%-2.8%) for major bleeding (Figure 2). When examining the ACS and CCS

subgroups, HFH consistently showed a higher frequency relative to other nonfatal adverse events (Supplemental Figures 2A and 2B).

CRUDE MORTALITY AFTER NONFATAL ADVERSE CLINICAL EVENTS.

Among the entire cohort, patients experiencing HFH had the highest crude mortality rate at 25.4% (122 of 481; 95% CI: 21.6%-29.5%), followed by those with major bleeding (9.3% [24 of 259]; 95% CI: 6.1%-13.6%) and ACS (8.8% [31 of 353]; 95% CI: 6.1%-12.4%). In contrast, patients without events showed a lower mortality rate of 3.6% (345 of 9,461; 95% CI: 3.3%-4.0%) (Table 2). The corresponding annualized mortality rates were highest for HFH (24.5%; 95% CI: 20.3%-29.2%), followed by major bleeding (8.5%; 95% CI: 5.5%-12.7%), and ACS (7.9%; 95% CI: 5.4%-11.2%). Similar trends were observed within the ACS and CCS cohorts, with HFH consistently demonstrating the highest mortality rates (ACS, 23.7% [95% CI: 18.3%-30.3%]; CCS, 25.3% [95% CI: 19.2%-32.8%]). In the unadjusted model (model 0), HFH showed the strongest association with mortality (HR: 13.64; 95% CI: 11.08-16.80; $P < 0.001$), followed by ACS (HR: 3.63; 95% CI: 2.51-5.29; $P < 0.001$) and major bleeding (HR: 3.54; 95% CI: 2.34-5.35; $P < 0.001$).

MULTIVARIABLE ASSOCIATION BETWEEN NONFATAL ADVERSE CLINICAL EVENTS AND SUBSEQUENT MORTALITY. In the age- and sex-adjusted model 1, HFH demonstrated the strongest association with mortality, followed by ACS and major bleeding (Table 3). These associations remained robust in model 3, with

TABLE 1 Baseline Characteristics of the Study Population

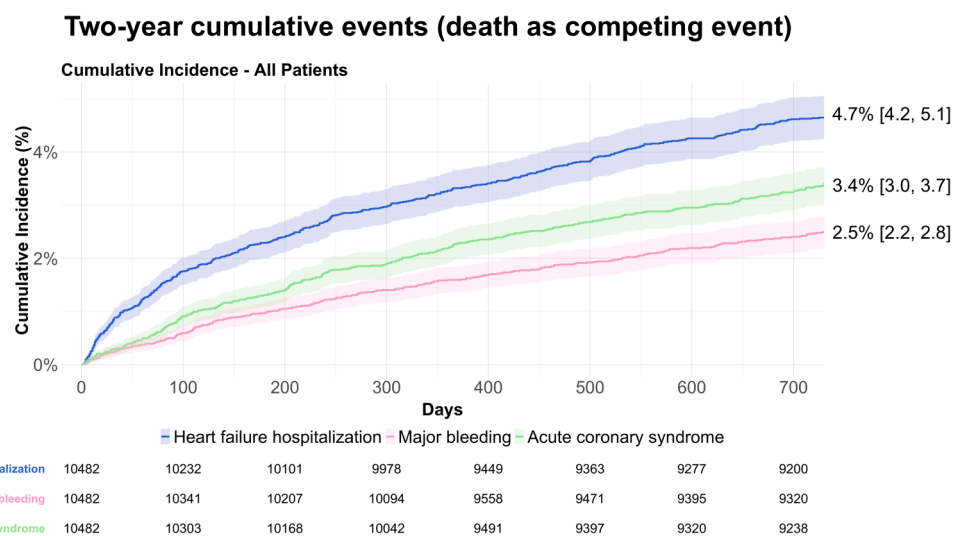
	Full Cohort (n = 10,482)	ACS Cohort (n = 5,401)	CCS Cohort (n = 5,081)
Patient characteristics			
Age, y	68.7 ± 11.4	67.4 ± 12.4	70.0 ± 10.0
Male	8,120 (77.5)	4,191 (77.6)	3,929 (77.3)
Medical history			
Myocardial infarction	1,556 (14.8)	577 (10.7)	979 (19.3)
Heart failure	853 (8.1)	244 (4.5)	609 (12.0)
Peripheral vascular disease	916 (8.7)	272 (5.0)	644 (12.7)
Cerebrovascular event	913 (8.7)	416 (7.7)	497 (9.8)
Hypertension	7,788 (74.3)	3,836 (71.0)	3,952 (77.8)
Diabetes	4,035 (38.5)	1,828 (33.8)	2,207 (43.4)
Dyslipidemia	6,547 (62.5)	3,211 (59.5)	3,336 (65.7)
Chronic kidney disease	2,010 (19.2)	995 (18.4)	1,015 (20.0)
Previous percutaneous coronary intervention	1,928 (18.4)	672 (12.4)	1,256 (24.7)
Previous coronary artery bypass grafting	486 (4.6)	145 (2.7)	341 (6.7)
Body mass index, kg/m ²	24.2 ± 3.6	24.0 ± 3.7	24.3 ± 3.6
Smoking	3,442 (32.8)	2,170 (40.2)	1,272 (25.0)
Clinical presentation			
STEMI	2,779 (26.5)	2,779 (51.5)	0 (0.0)
NSTEMI/unstable angina	2,622 (25.0)	2,622 (48.5)	0 (0.0)
Chronic coronary syndrome	5,081 (48.5)	0 (0.0)	5,081 (100.0)
Heart failure on admission	1,219 (11.6)	682 (12.6)	537 (10.6)
Cardiogenic shock	324 (3.1)	317 (5.9)	7 (0.1)
Cardiac arrest	200 (1.9)	191 (3.5)	9 (0.2)
Intra-aortic balloon pump	537 (5.1)	479 (8.9)	58 (1.1)
ECMO	28 (0.3)	24 (0.4)	4 (0.1)
Procedural characteristics			
Access site			
Radial	4,762 (45.4)	2,778 (51.4)	1,984 (39.0)
Femoral	5,585 (53.3)	2,581 (47.8)	3,004 (59.1)
Brachial	125 (1.2)	37 (0.7)	88 (1.7)
Target vessel number			
1	9,461 (90.3)	4,903 (90.8)	4,558 (89.7)
2	935 (8.9)	450 (8.3)	485 (9.5)
3	83 (0.8)	47 (0.9)	36 (0.7)
Lesion location			
Left anterior descending coronary artery	5,513 (52.6)	2,742 (50.8)	2,771 (54.5)
Right coronary artery	3,126 (29.8)	1,751 (32.4)	1,375 (27.1)
Left circumflex coronary artery	1,671 (15.9)	834 (15.4)	837 (16.5)
Left main trunk	387 (3.7)	160 (3.0)	227 (4.5)
Medication after PCI			
Dual antiplatelet therapy	9,854 (94.0)	5,073 (93.9)	4,781 (94.1)
β-blocker	7,238 (69.1)	4,195 (77.7)	3,043 (59.9)
Renin-angiotensin system inhibitor	6,702 (63.9)	3,882 (71.9)	2,820 (55.5)
Statin	8,952 (85.4)	4,828 (89.4)	4,124 (81.2)

Values are mean ± SD or n (%).

ACS = acute coronary syndrome(s); BMI = body mass index; CCS = chronic coronary syndrome; ECMO = extracorporeal membrane oxygenation; NSTEMI = non-ST-segment elevation myocardial infarction; PCI = percutaneous coronary intervention; STEMI = ST-segment elevation myocardial infarction.

HFH remaining the highest risk (aHR: 6.11; 95% CI: 4.76-7.85; $P < 0.001$), followed by ACS (aHR: 3.22; 95% CI: 2.14-4.84; $P < 0.001$) and major bleeding (aHR: 2.62; 95% CI: 1.71-4.02; $P < 0.001$). No significant interactions were observed among the adverse events (HFH vs major bleeding, P for interaction = 0.87; HFH vs ACS, P for interaction = 0.53; ACS vs major bleeding, P for interaction = 0.98). Furthermore, the prognostic impact of these events did not significantly differ between the ACS and CCS cohorts (P for interaction = 0.55 for HFH, 0.78 for ACS, and 0.26 for major bleeding).

Sensitivity analyses modeling all 3 adverse events simultaneously demonstrated results consistent with the primary analysis, with HFH maintaining the strongest association with all-cause mortality (Supplemental Table 3). Furthermore, the results remained robust after multiple imputation for missing data, demonstrating HRs consistent with the primary complete-case analysis (Supplemental Table 4). Stratification by heart failure history revealed a significant interaction with HFH (P for interaction = 0.002). Specifically, the mortality risk associated with HFH was more pronounced in

FIGURE 2 Cumulative Incidence of Heart Failure Hospitalization, Bleeding, and Acute Coronary Syndrome With Competing Risk for Death

Two-year cumulative incidence of heart failure hospitalization (4.7%; 95% CI: 4.2%-5.1%), major bleeding (2.5%; 95% CI: 2.2%-2.8%), and acute coronary syndrome (3.4%; 95% CI: 3.0%-3.7%), accounting for the competing risk for death. Shaded areas represent 95% CIs.

patients without histories of heart failure (aHR: 9.14; 95% CI: 6.36-13.13) compared with those with histories (aHR: 4.92; 95% CI: 3.55-6.82) ([Supplemental Table 5](#)). No significant interactions were observed for ACS or major bleeding (P for interaction = 0.16 and 0.49, respectively). Finally, in a sensitivity analysis restricted to events occurring within the first 180 days, HFH remained the strongest predictor of subsequent 1-year mortality ([Supplemental Table 6](#)).

In the fully adjusted model 3, the PAF for HFH was 20.1% (95% CI: 19.0%-21.0%), indicating that HFH

accounted for approximately one-fifth of the overall mortality burden. This contribution was substantially larger than that of ACS (4.3%; 95% CI: 3.3%-4.9%) or major bleeding (2.9%; 95% CI: 1.9%-3.5%) ([Central Illustration](#)). In the secondary analysis restricted to cardiovascular death, HFH and ACS showed substantial associations with the mortality burden ([Supplemental Figure 3](#)). HFH and ACS accounted for PAFs of 23.9% (95% CI: 21.8%-25.1%) and 11.0% (95% CI: 10.1%-11.4%), respectively. Conversely, the PAF for major bleeding was 1.8% (95% CI: -1.0% to 2.6%).

TABLE 2 Crude Association Between Incident Events and Mortality

	Heart Failure Hospitalization (n = 481)	Major Bleeding (n = 259)	Acute Coronary Syndrome (n = 353)	No Event (n = 9,461)
Full cohort (n = 10,482)				
All-cause mortality, n (%) [95% CI]	122 (25.4 [21.6-29.5])	24 (9.3 [6.1-13.6])	31 (8.8 [6.1-12.4])	345 (3.6 [3.3-4.0])
Annual mortality rate after the event, %/y (95% CI)	24.5 (20.3-29.2)	8.5 (5.5-12.7)	7.9 (5.4-11.2)	1.9 (1.7-2.1)
Total follow-up duration (postevent), person-years	498.8	281.3	391.9	18,109.4
ACS cohort (n = 5,401)				
All-cause mortality, n (%) [95% CI]	65 (26.1 [20.9-32.1])	9 (6.8 [3.4-12.9])	17 (7.5 [4.5-11.9])	165 (3.4 [2.9-4.0])
Annual mortality rate after the event, %/y (95% CI)	23.7 (18.3-30.3)	6.6 (3.0-12.5)	6.5 (3.8-10.4)	1.8 (1.5-2.1)
Total follow-up duration (postevent), person-years	273.8	136.7	261.5	9,231.7
CCS cohort (n = 5,081)				
All-cause mortality, n (%) [95% CI]	57 (24.6 [19.3-30.7])	15 (11.8 [7.0-19.0])	14 (11.2 [6.5-18.4])	180 (3.9 [3.4-4.5])
Annual mortality rate after the postevent, %/y(95% CI)	25.3 (19.2-32.8)	10.4 (5.8-17.1)	10.7 (5.9-18.0)	2.0 (1.7-2.3)
Total follow-up duration (postevent), person-years	225.0	144.6	130.5	8,877.7

Abbreviations as in [Table 1](#).

TABLE 3 Associations Between Incident Events and Mortality Using Cox Proportional Hazards Model in Separate Analyses

	Model 1 (n = 10,439)		Model 2 (n = 9,376)		Model 3 (n = 9,241)	
Full cohort (n = 10,482)						
Missing, n (%)	43 (0.4)		1,106 (10.6)		1,241 (11.8)	
	aHR (95% CI)	P Value	aHR (95% CI)	P Value	aHR (95% CI)	P Value
Heart failure hospitalization	10.05 (8.11-12.46)	<0.001	6.15 (4.81-7.87)	<0.001	6.11 (4.76-7.85)	<0.001
Major bleeding	2.75 (1.82-4.16)	<0.001	2.58 (1.68-3.94)	<0.001	2.62 (1.71-4.02)	<0.001
Acute coronary syndrome	3.67 (2.53-5.34)	<0.001	3.29 (2.22-4.89)	<0.001	3.22 (2.14-4.84)	<0.001
ACS cohort (n = 5,401)	(n = 5,397)		(n = 5,096)		(n = 5,021)	
Missing, n (%)	4 (0.1)		305 (5.6)		380 (7.0)	
	aHR (95% CI)	P Value	aHR (95% CI)	P Value	aHR (95% CI)	P Value
Heart failure hospitalization	10.00 (7.41-13.48)	<0.001	6.43 (4.58-9.02)	<0.001	6.19 (4.37-8.77)	<0.001
Major bleeding	2.34 (1.20-4.60)	0.01	1.85 (0.90-3.83)	0.10	1.85 (0.89-3.84)	0.10
Acute coronary syndrome	3.23 (1.94-5.37)	<0.001	3.06 (1.82-5.14)	<0.001	3.07 (1.80-5.24)	<0.001
CCS cohort (n = 5,081)	(n = 5,042)		(n = 4,280)		(n = 4,220)	
Missing, n (%)	39 (0.8)		801 (15.8)		861 (16.9)	
	aHR (95% CI)	P Value	aHR (95% CI)	P Value	aHR (95% CI)	P Value
Heart failure hospitalization	10.10 (7.41-13.76)	<0.001	5.80 (4.02-8.38)	<0.001	5.84 (4.03-8.47)	<0.001
Major bleeding	3.10 (1.83-5.25)	<0.001	3.11 (1.83-5.30)	<0.001	3.22 (1.89-5.50)	<0.001
Acute coronary syndrome	4.42 (2.54-7.68)	<0.001	3.56 (1.91-6.61)	<0.001	3.37 (1.76-6.45)	<0.001

Model 1 was adjusted for age and sex. Model 2 was adjusted for covariates in model 1 plus body mass index, estimated glomerular filtration rate, smoking status, hypertension, diabetes mellitus, dyslipidemia, previous myocardial infarction, previous heart failure, heart failure symptoms at index admission, peripheral vascular disease, cerebrovascular disease, previous percutaneous coronary intervention, and the indication for the index percutaneous coronary intervention. Model 3 was adjusted for covariates in model 2 plus oral medications after percutaneous coronary intervention. P values for interaction between ACS and CCS cohorts in model 3 were 0.55 for heart failure hospitalization, 0.26 for major bleeding, and 0.78 for acute coronary syndrome.

aHR = adjusted HR; other abbreviations as in Table 1.

DISCUSSION

In this multicenter PCI registry of 10,482 patients with a median follow-up of 2 years, nonfatal adverse events were not only common but also varied greatly in their contribution to long-term mortality risk. Notably, HFH emerged as the most frequent nonfatal adverse event, occurring in 4.7% of patients, followed by ACS at 3.4% and major bleeding at 2.5% over 2 years, with these trends consistent across both ACS and CCS subgroups. Multivariable analyses demonstrated that HFH was substantially associated with subsequent mortality, with fully adjusted HRs of approximately 6.11, while ACS and major bleeding were associated with HRs of 3.22 and 2.62, respectively. PAF analysis further emphasized HFH's clinical significance, revealing that it accounted for approximately one-fifth (20.1%) of the overall mortality burden, substantially higher than the contributions from ACS (4.3%) and major bleeding (2.9%). These findings underscore the critical prognostic implication of post-PCI nonfatal adverse events, particularly HFH, and highlight the need for targeted strategies to mitigate these complications in contemporary cardiovascular practice.

In our study, HFH emerged as the most frequent adverse event, with an incidence of approximately 5% over 2 years and a 6-fold increase in all-cause

mortality risk. This incidence is consistent with earlier studies, such as the CORONOR (Suivi d'Une Cohorte de Patients Coronariens Stables en Région Nord-pas-de-Calais) study, which revealed that approximately 6% of patients with coronary artery disease (CAD) developed heart failure within 5 years,¹⁷ and the CARE (Cholesterol and Recurrent Events) study, which reported a post-myocardial infarction heart failure incidence of 1.3% per year.¹⁸ Given the persistent heart failure-related vulnerability, even with contemporary improvements in revascularization techniques and optimal medical therapy, our findings highlight the need for targeted surveillance and aggressive risk factor control. Emerging therapies, including early initiation of sodium-glucose cotransporter 2 inhibitors after acute myocardial infarction, as demonstrated in DAPA-MI (Dapagliflozin Effects on Cardiometabolic Outcomes in Patients With an Acute Heart Attack) and EMPACT-MI (A Streamlined, Multicentre, Randomised, Parallel Group, Double-Blind Placebo-Controlled Superiority Trial to Evaluate the Effect of Empagliflozin on Hospitalisation for Heart Failure and Mortality in Patients With Acute Myocardial Infarction), although failing to show improvements in mortality rates, demonstrated modest beneficial effects of empagliflozin in preventing HFH.^{19,20} These agents may still provide clinical value for select CAD patients.

CENTRAL ILLUSTRATION Population Attributable Fractions of Post-PCI Events for All-Cause Mortality**Relative Impact of Heart Failure Hospitalization, Acute Coronary Syndrome, and Major Bleeding on Subsequent Mortality After PCI, N = 10,482****Study Design**

- 
- KiCS PCI Registry**
- 10,482 patients undergoing PCI (48% for ACS)
 - 15 centers in Tokyo metropolitan area
 - Mean 69 yrs, Women 22%
 - From 2008 to 2021
 - Median follow-up period 2 years

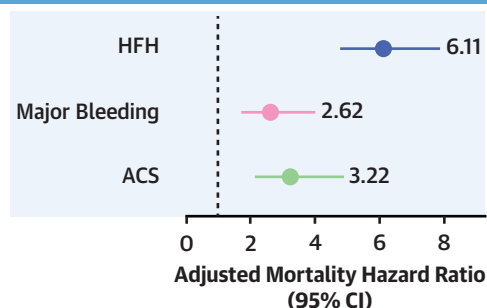
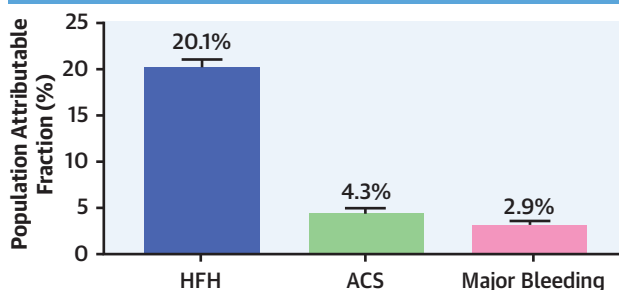
Primary outcome:

All-cause mortality

Exposures of interest:

Post-PCI nonfatal adverse events

- Heart failure hospitalization (HFH)
- Major bleeding
- Acute coronary syndrome (ACS)

Association Between Nonfatal Adverse Clinical Events and Subsequent Mortality**Population Attributable Fractions: Contribution of Post-PCI Events to Overall Mortality**

- In patients undergoing PCI, HFH was more strongly associated with subsequent mortality than ACS or major bleeding.
- 2-Year cumulative incidence rates were 4.7% (95% CI: 4.2-5.1) for HFH, 3.4% (3.0-3.7) for ACS, and 2.5% (2.2-2.8) for major bleeding.
- Patients with HFH post-PCI should be considered at high risk and needs enhanced follow-up and intervention strategies.

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The population attributable fractions for heart failure hospitalization (HFH), acute coronary syndrome (ACS), and major bleeding were 20.1% (95% CI: 19.0%-21.0%), 4.3% (95% CI: 3.3%-4.9%), and 2.9% (95% CI: 1.9%-3.5%), respectively, in model 3. PCI = percutaneous coronary intervention.

Additionally, a structured heart failure-focused management pathway in post-PCI care may mitigate the risk. Identifying high-risk subgroups (eg, diabetes, renal dysfunction, or a history of heart failure) could facilitate closer follow-up and earlier interventions. Future strategies should integrate heart failure prevention with ischemia and bleeding management, fostering a more personalized approach to CAD care.

ACS and major bleeding demonstrated comparable mortality risks, suggesting similar prognostic implications of these events. This finding aligns with large-scale systematic reviews that have shown the mortality risk associated with major bleeding to be equivalent to that of recurrent myocardial infarction.²¹ Despite advancements in stent design (eg, drug-eluting stents) and antithrombotic regimens, both bleeding and recurrent ischemia continue

to be major drivers of late morbidity and mortality. Recent clinical trials, including STOPDAPT-2 (Short and Optimal Duration of Dual Antiplatelet Therapy-2), STOPDAPT-3 (Short and Optimal Duration of Dual Antiplatelet Therapy-3), and MASTER DAPT (Management of High Bleeding Risk Patients Post Bioresorbable Polymer Coated Stent Implantation With an Abbreviated Versus Prolonged DAPT Regimen), have attempted to optimize the duration of dual antiplatelet therapy to balance ischemic and bleeding risks.²²⁻²⁴ Current guidelines recommend prioritizing bleeding risk assessment, which is particularly relevant given our findings of substantial prognostic significance of bleeding events in CCS patients with traditionally low thrombotic risk. Moreover, trials such as ISCHEMIA and COURAGE (Clinical Outcomes Utilizing Revascularization and Aggressive Drug Evaluation) in CCS patients, along

with REVIVED-BCIS (Revascularisation for Ischaemic Ventricular Dysfunction: A Randomized Comparison of Percutaneous Coronary Intervention [With Optimal Medical Therapy] Versus Optimal Medical Therapy Alone for Treatment of Heart Failure Secondary to Coronary Disease) in those with more advanced left ventricular dysfunction, demonstrate that optimal medical therapy remains crucial for long-term mortality, even after revascularization.^{5,25,26} Our data re-emphasize the clinical gravity of both bleeding and ischemic recurrence and highlight the need for careful evaluation of optimal dual antiplatelet therapy duration and vigilant monitoring in accordance with each patient's risk profile.

In contemporary cardiovascular clinical trials, composite endpoints are frequently used to capture multiple clinically relevant events under a single outcome measure.^{27,28} Although this approach enhances statistical power and feasibility by aggregating event rates, thus reducing the required sample size or follow-up duration, it poses challenges in evaluating events with differing clinical severity and pathophysiological mechanisms. This limitation, which may obscure crucial prognostic signals, has been recognized in consensus statements.²⁹ Our findings, particularly the disproportionately higher mortality risk associated with HFH, highlight this issue. To address such limitations, novel analytical approaches such as the win ratio have been introduced to hierarchically prioritize critical events, ensuring that outcomes of greater clinical significance carry appropriate weight.³⁰ Given the gradient of association between commonly incorporated endpoints (HFH, ACS, and bleeding) and mortality in our study, a more nuanced assessment of treatment effects across multiple outcomes is warranted. Integrating advanced methodologies while maintaining transparency in endpoint definitions³¹ and ensuring standardization within composite frameworks will be essential.³² As strategies to mitigate adverse outcomes post-PCI, these refinements will help optimize patient care and enhance the clinical applicability of trial findings.

STUDY LIMITATIONS. First, although we adjusted for multiple confounders in our analyses, residual confounding may exist because of unmeasured variables or changes in medical therapy during follow-up. For instance, we lacked detailed information about medication adherence and modifications in medical therapy following post-PCI events. Regarding baseline cardiac function, although quantitative echocardiographic data (eg, left ventricular ejection fraction) and valvular disease status were unavailable, we adjusted for “history of heart failure” and

“heart failure symptoms on admission,” which serve as clinical proxies for baseline cardiac function in our analysis. These unmeasured parameters could influence both the occurrence of adverse events and subsequent mortality risk.

Second, this study was conducted using data from selected hospitals within the Tokyo metropolitan area and surrounding prefectures, which may limit the generalizability of our findings to other geographic regions or health care settings with different patient characteristics or practice patterns.

Third, although we implemented a comprehensive follow-up system, some clinical events might have been missed if patients were treated at nonparticipating hospitals. This potential underreporting of events could have affected our estimates of cumulative incidence.

Fourth, we were unable to evaluate the association between adverse events and cardiovascular mortality because of the inability to determine the exact cause of death. However, by focusing on all-cause mortality, we minimized the risk for misclassification and enhanced the accuracy of our data.

Fifth, our study was focused on patients who survived their index hospitalization for PCI, which may have introduced survival bias by excluding early adverse events occurring during the initial hospitalization. Additionally, the follow-up period was limited to 2 years. Consequently, for patients who developed adverse events near the end of the follow-up period, the remaining observation period was short. This may have led to an underestimation of the association between these events and subsequent mortality, suggesting that a longer follow-up would be of interest.

Finally, regarding the PAF analysis, we acknowledge that it relies on the assumption of a relationship between the exposure and outcome. Accordingly, the PAFs reported in this study should be interpreted as reflecting the proportion of all-cause mortality statistically associated with post-PCI nonfatal events, rather than the proportion of deaths that would be strictly preventable by eliminating these exposures. Our intent was to quantify the relative population-level mortality burden associated with HFH, ACS, and major bleeding in a real-world PCI population rather than to make definitive causal claims.

CONCLUSIONS

In this large, multicenter PCI registry, we demonstrate that nonfatal adverse events were significantly associated with subsequent mortality, with HFH showing the strongest association. The present study

suggests that post-PCI patients who experience HFH should be considered at exceptionally high risk and may benefit from enhanced follow-up and intervention strategies. Proactive measures, including closer monitoring and optimizing heart failure management, could potentially improve survival outcomes in this vulnerable group.

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PERSPECTIVES

WHAT IS KNOWN? Current guidelines and quality metrics commonly assign equal weight to nonfatal adverse events such as HFH, ACS, and major bleeding within composite endpoints following PCI, despite limited evidence comparing their relative prognostic significance.

WHAT IS NEW? HFH was associated with substantially higher mortality risk (aHR: 6.11) than ACS (aHR: 3.22) or major bleeding (aHR: 2.62) and accounted for approximately 20% of the overall mortality burden compared with approximately 4% and 3% for the other events, respectively.

WHAT IS NEXT? Future trials should consider endpoint strategies that account for clinical priority and heterogeneous prognostic severity across nonfatal events, and post-PCI care pathways should prioritize heart failure surveillance and prevention strategies in high-risk patients.

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APPENDIX For supplemental figures and tables, please see the online version of this paper.